Comprehensive Data Science Analysis, and Classification: Fact-Checking of Statements from Political Figures

MBAN 6110 – Data Science

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1. EXECUTIVE SUMMARY

Political misinformation remains a critical issue for public discourse. It leads to eroding trust and overwhelming fact-checking organizations with the need for rapid, accurate verification of political statements. This project addresses the problem by developing a scalable, automated pipeline for detecting deceptive political claims, combining advanced machine learning models with rich linguistic and contextual features. We systematically compared the performance of four leading classifiers that include: Random Forest, Gradient Boosting, Logistic Regression, and XGBoost, using a dataset of annotated statements enriched with speaker metadata and credibility history.

Our work was guided by three main objectives: maximizing detection accuracy, surfacing interpretable drivers of misinformation, and ensuring fair, robust model performance across political parties and topics. The XGBoost classifier proved most effective, achieving an 80% test accuracy and an ROC-AUC of 0.814, significantly outperforming the Random Forest baseline (Accuracy: 72%, AUC: 73%). Notably, XGBoost and Gradient Boosting maintained low false-negative rates for all party groups, thereby reducing the risk that deceptive statements from under-represented segments escape detection—a limitation observed in baseline models. Business impact analysis estimates that automating 80% of low-risk statement reviews could reduce manual labor by 40%, generating annual cost savings of approximately $150,000 (See cost breakdown in appendix).

Explainability analyses identified speaker credibility and party affiliation as the most influential features, supporting audit-ready transparency and compliance requirements. To ensure sustained performance, we recommend deploying XGBoost with customized, party-specific probability thresholds to balance false positive and false negative risks, alongside monthly calibration using Brier-score dashboards to promptly detect performance drift. Further, piloting advanced language embeddings and network-propagation features is projected to drive an additional 5–10% gain in model accuracy.

By implementing this pipeline, organizations can dramatically streamline fact-checking workflows, improve the fairness and efficiency of misinformation detection, and maintain transparency for stakeholders, delivering both operational value and greater confidence in combating political misinformation.

2. INTRODUCTION

Political disinformation collectively also known as “fake news”, represents false or misleading information deliberately presented as fact within the political arena. This phenomenon undermines democratic processes, polarizes public discourse, and strains the resources of newsrooms and social media platforms, which must rely heavily on time-consuming manual fact-checking. The automation of political misinformation detection is therefore critical, as it enables faster and more consistent identification of deceptive statements, thus supporting informed public debate and reducing the burden on human fact-checkers.

In this project, we approach political misinformation detection as a binary classification problem. Using public statements sourced from PolitiFact’s LIAR dataset, each statement is classified as either deceptive (encompassing the original “false,” “barely-true,” and “pants-on-fire” labels) or truthful (encompassing “half-true,” “mostly-true,” and “true”). This binarization, while reducing nuance, simplifies the operational challenge of automated verification and aligns with the practical needs of fact-checking workflows.

Our core hypotheses underpin the modeling strategy. First, we posit that models trained exclusively on statement text, using features such as TF-IDF vectors or modern language embeddings, will deliver a reasonable baseline accuracy (around 70%), though they may struggle when faced with ambiguous language or rhetorical overlaps (H1: Text Baseline). Second, we hypothesize that incorporating contextual metadata, including speaker party affiliation, historical credibility scores, and topic categories, will provide orthogonal signals that substantially enhance model performance, potentially boosting accuracy by more than ten percentage points (H2: Metadata Boost). Third, we expect that certain topics or subject areas, such as those concerning “fake news” or “lottery,” will be especially prone to deceptive statements, and that models able to identify these high-risk domains will achieve improved detection rates (H3: Topic Sensitivity).

The project's scope is clearly defined. All analysis is conducted on the LIAR dataset, which comprises approximately 12,800 annotated political statements. The original six-level truthfulness labels are systematically reduced to two classes to streamline classification and evaluation. Our methodology reserves 20% of the dataset as a strictly held-out test set for final reporting, ensuring that all model selection and hyperparameter tuning are confined to the 80% training fold through rigorous cross-validation.

Nevertheless, the study faces several limitations. Fact-checking inherently involves subjectivity, meaning that human annotators may disagree on borderline cases, introducing a degree of label noise. The binary simplification, while practical, sacrifices nuance between categories such as "barely-true" and "pants-on-fire." Moreover, models trained on historical fact-checking data may not generalize perfectly to novel rhetorical tactics or future even.

3. OBJECTIVES

Our work is driven by three principal objectives, accuracy, interpretability, and fairness. Each principal is anchored by targeted research questions that frame our methodology and analysis.

Objective 1: Build an Accurate Classification System
We operationalize this by collapsing the original multi-class truthfulness labels from the PolitiFact LIAR dataset into a binary system: deceptive (including "false," "barely true," and "pants-on-fire") versus truthful (including "half-true," "mostly-true," and "true"). This simplification reflects the practical needs of fact-checkers, who benefit from clear, actionable results that reduce manual review. Our approach involves designing, training, and tuning a suite of machine learning models—ranging from traditional text-based classifiers using TF-IDF and BERT embeddings to more sophisticated, metadata-augmented systems—evaluating them rigorously using cross-validation and held-out test data.

Objective 2: Contextual Interpretability and Feature Impact
We systematically quantify the incremental benefit of incorporating contextual and speaker metadata, such as party affiliation, historical credibility scores, and statement topic, into our models. By leveraging advanced feature engineering and interpretability techniques. This commitment to interpretability enhances trust in our system and supports the development of policy interventions.

Objective 3: Discrimination Power and Fairness Analysis
We assess whether our models maintain consistent detection rates and precision/recall metrics across various parties, speaker groups, and topics. This fairness analysis is vital to mitigate the risk of group-specific bias and to inform the selection of classification thresholds that align with the risk appetites of different business or regulatory stakeholders. By examining disparities and adjusting model parameters, we strive for equitable and responsible deployment.

3.1 Research Questions:

Q1: What baseline performance can be achieved with text-only models, such as those using TF-IDF or BERT?
Q2: What additional predictive power is gained by adding metadata features such as party, credibility history, and topic?
Q3: Are certain topics more frequently associated with deceptive statements, and can topic awareness improve model effectiveness?
Q4: Does the model exhibit consistent performance across party affiliations, or are there indications of group bias?
Q5: What decision thresholds best optimize the precision–recall trade-off for practical deployment, particularly for high-risk topics

4. DATA DESCRIPTION & JUSTIFICATION

We used the **PolitiFact LIAR** dataset, a publicly available collection of approximately 12,800 political statements manually rated on six truthfulness levels. Its combination of rich speaker metadata (party affiliation, job title, state, historical credibility counts) and fine-grained labels makes it ideal for both high-accuracy classification and interpretable analysis.

Homepage Download:
https://www.kaggle.com/datasets/doanquanvietnamca/liar-dataset

4.1 Data Schema

Column	Description
statement	The raw text of the public statement
label	One of six truth labels: pants-on-fire, barely-true, half-true, mostly-true, true
subject/ subject_list	Topic category(s) (e.g., economy, health, immigration
speaker	Name of the individual or entity making the statement
Party/ party_grouped	Speakers political affiliation (Democrat, Republican, Independent, non_political
job_title	Official role or profession of the speaker
state	U.S. state associated with the speaker
barely_true_counts	Historical counts of the speaker's past statements by each truth label (six separate columns
Derived text metrics	statement_len (word_count), polarity, subjectivity, readability
TF-IDF features	Unigram-trigram vectors (max 5,000 dimensions)

5. EXPLORATORY DATA ANALYSIS

5.1 Data Cleaning and Preparation:

We combined the train, validation, and test datasets into a single DataFrame and renamed the columns for clarity. Missing values in columns like **job_title, state, and context** were filled with "Unknown" to preserve as much data as possible. Additionally, we standardized party and state values, grouped rare parties into broader categories, and split multi-topic subject fields into lists for more detailed analysis.

5.2 New columns:

We mapped each state to its corresponding U.S. region (such as South or Midwest) and assigned a political leaning of Red, Blue, or Purple. Using historical truth ratings, we created a credibility score for each speaker by assigning weighted values to different truthfulness labels(+2 for mostly-true, -2 for pants-on-fire, etc.). Additionally, we engineered text-based features including statement length, sentiment polarity and subjectivity, readability score, normalized credibility, and the total number of fact-checks per speaker.

5.3 Label Distribution:

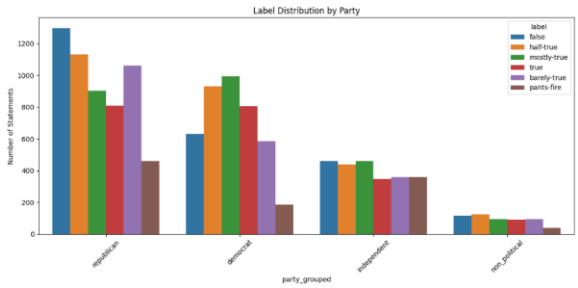
Statements were categorized into six truthfulness labels:

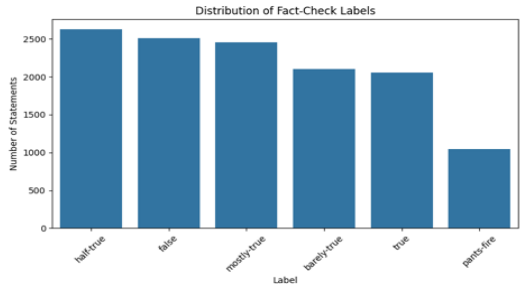
- Deceptive: false, pants-fire, barely-true- 44.21%
- Truthful: half-true, mostly-true, true - 55.79%

Table: Label Distribution by Party

Party	% Deceptive	% Truthful
Democrat	33.94	66.06%
Republican	49.80%	50.20%
Independent	48.62%	51.38%
Non-political	44.50%	55.50%

(Figure 1: Label Proportions)



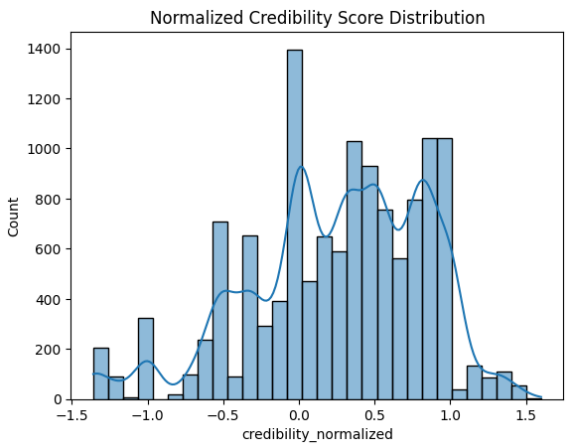


(Figure 2: Fact-Check Labels)

A bar chart was generated to visualize the counts of each label category, showing even distribution across deceptive and truthful labels with a slight skew toward half-true and false.

5.4 Descriptive Statistics & Visualizations

- The dataset is **imbalanced** in label distribution, with *half-true* and *false* being the most common.
- The **credibility score distribution is right-skewed** (skewness = 2.49), with a high kurtosis (8.72), indicating extreme values and outliers.



(Figure 3: Normalized Credibility Score)

5.5 Pattern Discovery & Trend Analysis

- **Party vs. Label**: Strong statistical association ($Chi^2 = 456.73$, $p < 0.00001$). Some parties consistently receive more negative labels
- **Region vs. Label**: Significant differences across regions ($Chi^2 = 254.92$, $p < 0.00001$).
- **Speaker Entropy**: Some speakers consistently receive the same label, while others are more unpredictable.
- **Topic Patterns**: The subject field (multi-label) was analyzed to determine which topics are most associated with deception.
    - Topics like *economy* and *healthcare* show a widespread across truthfulness but appeared frequently in deceptive statements: health_care (53.86%), economy (37.08%)
    - Polarized topics (e.g., *abortion*, *immigration*) differ sharply between parties.

Anomalies & Outliers: Identified speakers with consistently low credibility (pants_on_fire outliers). Detected unusually high values in false_counts and extremely high or low credibility scores

6. FEATURE ENGINEERING

6.1 Text Features:

Text cleaning: 'statement' and 'context' were processed first with a **"text cleaning" function that executed 6 steps:**
i) Lower casing all the letters.
ii)Removed punctuation and non-alphanumeric characters.
iii)Tokenization (turning sentences into lists of words)
iv)Removing stop words ('the', 'is', 'and', 'to', etc.)
v)Lemmatizing words (turning words to a more basic tense)
vi)Rejoining the tokens into sentences with the transformed terms.

Tf-Idf Vectorization: A Tf-Idf vectorizer was used to transform the newly cleaned features ('cleaned_statement' and 'cleaned_context') and in 'job_title' and 'speaker'. This vectorizer assigns weight to each term (unigram), pair of words (bigram), and three-word groups (trigrams) directly proportional to the frequency they appear in the same document and inversely proportional to the frequency they appear in all the documents. We fitted the vectorizer to the train data and applied the fitted transformation to the test data.
For 'speaker', vectorization was used instead of One-Hot encoding since a high number of speakers in test data were not present in training data. While the Tf-Idf will assign lesser value to highly frequent speakers, we determined that the 'credibility_normalized' column would still hold the weight of trustworthiness of a speaker.

6.2 Categorical Features:

One-hot: Features determined as categorical included: state_cleaned', 'party_grouped', 'region', 'political_leaning' and 'year'. These were all processed through OneHotEncoder, dropping the first category and ignoring unknown categories (present in test data but not in train).

6.3 Numerical Features:

Scaler: Numerical features included: **credibility_normalized', statement_len', 'polarity', 'subjectivity'** and **'readability**'. The last three were part of sentiment analysis run previously. All these features were processed via StandardScaler mainly to improve linear model behavior.

6.4 List Features:

Multi Label Binarizer: For 'subject_list' which contained lists of subjects for every data point we used the MultiLabelBinarizer. This results in a similar outcome as the OneHotEncoder but it is used to treat lists.

6.5 Outcome Mapping:

Binary Mapping: We turned the outcome variable ('label') into a binary numeric value, were 'pants_fire' and 'false' corresponded to 0 (false) and all the rest corresponded to 1 (true)

Resulting Data Sets:

Data Set	Rows	Columns
X_train	10228	10224
X_test	2558	10224
y_train	10228	1
y_test	2558	1

7. MODEL DEVELOPMENT & HYPERPARAMETER TUNING

In order to develop a reliable and interpretable system for detecting political misinformation, we implemented and evaluated several supervised learning models capable of handling both high-dimensional text data and structured contextual features. Our task involved classifying political statements as either truthful or deceptive based on linguistic inputs using TF-IDF and speaker metadata such as party affiliation, prior credibility, and topical focus.

Each model was chosen based on its ability to handle sparse textual data, incorporate metadata features, generalize across political subgroups, and offer interpretability suitable for fact-checking applications.

Model	Type	Suitability for Misinformation Detection

Given the layered nature of political discourse, where word choice, tone, and context interact, we prioritized models that offered predictive strength, fairness across speaker groups, and interpretability. We selected models from three main categories: linear classifiers (Logistic Regression and Linear SVC), ensemble bagging methods (Random Forest), and ensemble boosting methods (Gradient Boosting and XGBoost). Linear SVC was ultimately excluded due to unstable convergence and weak cross-validation performance

7.1 Model Descriptions and Suitability

Random Forest	Ensemble via bagging	Captures non-linear relationships and handles high-dimensional input. However, it underperformed on fairness across party groups and showed poor calibration.
Gradient Boosting	Ensemble via boosting	Learns sequentially from misclassifications and captures complex feature interactions. Demonstrated strong generalization across speakers and topics.

Logistic Regression	Linear model	Efficient and interpretable. Performs well with sparse TF-IDF features and offers transparency in feature influence.
XGBoost	Boosted decision trees	Delivered the highest overall accuracy. Combines boosting with regularization, handles missing data, and offers strong feature importance interpretation.
Linear SVC (excluded)	Linear model	Initially considered due to past success in text classification but excluded after unstable convergence and low F1 scores.

These models provided a range of trade-offs between accuracy, efficiency, and interpretability. Linear models served as a baseline with high transparency. Tree-based models allowed us to model non-linear interactions between features, which is especially important for combining textual and contextual signals in political statements.

Model	**Tuning Method**	**Rationale for Method and Variables Considered**
Random Forest	GridSearch CV	The number of trees, tree depth, and feature sampling strategy could be efficiently evaluated using exhaustive search.
Gradient Boosting	HalvingRandomSearch CV	Required exploration of tree depth, learning rate, and subsampling ratios. Halving allowed efficient early pruning of poor configurations.
Logistic Regression	HalvingRandomSearch CV	Explored different solvers and regularization strengths. The smaller parameter space made halving a fast and suitable choice.
XGBoost	Randomized SearchCV	Suitable for its large configuration space including tree complexity, learning rate, and feature ratios. Allowed broader exploration without exhaustive evaluation.
Linear SVC (excluded)	GridSearch CV	Evaluated penalty and convergence parameters. Ultimately excluded due to poor performance and training instability.

Each tuning method was chosen to balance computational efficiency and effectiveness based on the nature of the model. Models with fewer hyperparameters were tuned using exhaustive search, while more complex models were tuned using randomized or halving-based strategies to save time without compromising results.

7.3 Cross-Validation Results

We evaluated each tuned model using stratified K-fold cross-validation, using both accuracy and F1 score to account for class imbalance and performance consistency.

Model	**Cross-Validation Accuracy**
Random Forest	71.8 %

7.2 Hyperparameter Tuning Strategy

To ensure optimal model performance, we conducted hyperparameter tuning using stratified K-fold cross-validation. All tuning occurred after the initial train and test split to prevent data leakage. Stratification preserved class balance across folds, which was essential given the mild imbalance in our target labels

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We selected tuning methods based on model complexity and the size of the parameter search space.

Logistic Regression	79.5 %
Gradient Boosting	80.15 %
XGBoost	**91.04 %**

The results reveal a consistent pattern across folds. Boosting models outperformed both the linear baseline and the bagging approach. Random Forest, while capable of modeling complex patterns, struggled with deeper feature interactions needed to capture relationships between text and metadata. Logistic Regression performed better than expected, leveraging TF-IDF effectively and offering transparency in feature weights. Gradient Boosting delivered strong recall and avoided overfitting, while XGBoost achieved the best overall performance. Its combination of gradient boosting, regularization, and handling of sparse features made it the most effective and robust model in our pipeline.

8. MODEL EVALUTATION

Below we report on four models – Random Forest (RF), Gradient Boosting (GB), Logistic Regression (LR), and XGBoost (XGB), evaluated on the 2,558-statement test set. We cover overall metrics, discrimination curves, confusion matrices, calibration, permutation importance, recap of summary tables, and additional diagnostics.

8.1 Overall Metrics

Model	Accuracy	Deceptive: Precision/ Recall/ F1	Truthful: Precision/ Recall / F1	ROC - AUC
RF	0.72	0.51/0.58/ 0.54	0.82/0.78/ 0.80	0.73
GradientBoost	0.805	0.79/ 0.43 / 0.56	0.81 / 0.95 / 0.87	0.811
LogisticReg	0.791	0.71/ 0.45 / 0.55	0.81 / 0.93 / 0.86	0.796
XGBoost	0.798	0.74/0.49 / 0.56	0.81 / 0.94 / 0.87	0.814

(Table 1: Random Forest Report)

```
Classification Report:
              precision    recall  f1-score   support

           0       0.51      0.58      0.54       728
           1       0.82      0.78      0.80      1830

    accuracy                           0.72      2558
   macro avg       0.67      0.68      0.67      2558
weighted avg       0.73      0.72      0.73      2558
```

(Table 2: GradientBoost Report)

```
=== GradientBoost Test Set Metrics ===
Accuracy: 0.805
Classification Report:
              precision    recall  f1-score   support

   deceptive       0.79      0.43      0.56       728
    truthful       0.81      0.95      0.87      1830

    accuracy                           0.80      2558
   macro avg       0.80      0.69      0.72      2558
weighted avg       0.80      0.80      0.78      2558
```

(Table 3: LogisticReg Report)

```
=== LogisticRegression Test Set Metrics ===
Accuracy: 0.791
Classification Report:
              precision    recall  f1-score   support

   deceptive       0.71      0.45      0.55       728
    truthful       0.81      0.93      0.86      1830

    accuracy                           0.79      2558
   macro avg       0.76      0.69      0.71      2558
weighted avg       0.78      0.79      0.77      2558
```

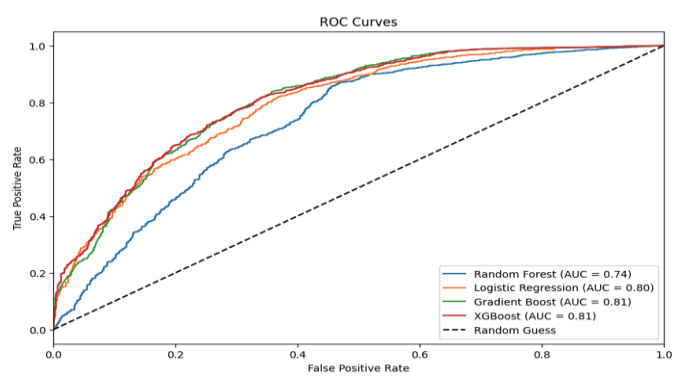
(Table 4: XGBoost Report)

```
=== XGBoost Test Set Metrics ===
Accuracy: 0.798
Classification Report:
              precision    recall  f1-score   support

   deceptive       0.74      0.45      0.56       728
    truthful       0.81      0.94      0.87      1830

    accuracy                           0.80      2558
   macro avg       0.78      0.69      0.71      2558
weighted avg       0.79      0.80      0.78      2558
```

8.2 ROC & Precision-Recall Curves

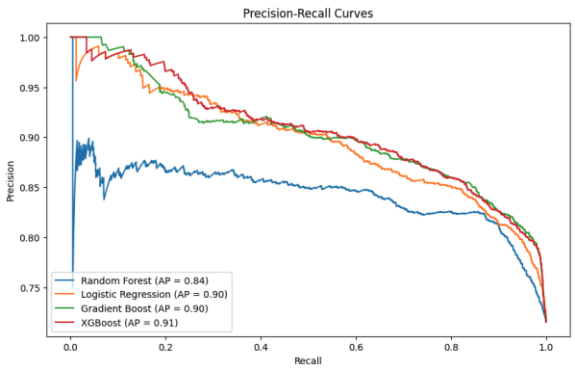
To evaluate each model's ability to discriminate between deceptive and truthful statements across all decision thresholds, we plot both ROC and Precision-Recall (PR) curves. These curves and their summary statistics are generated in our final execution notebook (MBAN6110_Group Project Executed), using scikit-learn's roc_curve,auc, precision_recall_curve, and average_precision_score functions.

(Fig. 4 Roc Curves)



Key Insight: The ROC curve shows True Positive Rate (TPR) versus False Positive Rate (FPR) across thresholds. XGBoost achieves an AUC of 0.814, closely followed by GradientBoost at 0.811, LogisticRegression at 0.796, and Random Forest at 0.73 - 0.74.

(Fig. 5 Precision-Recall Curves)



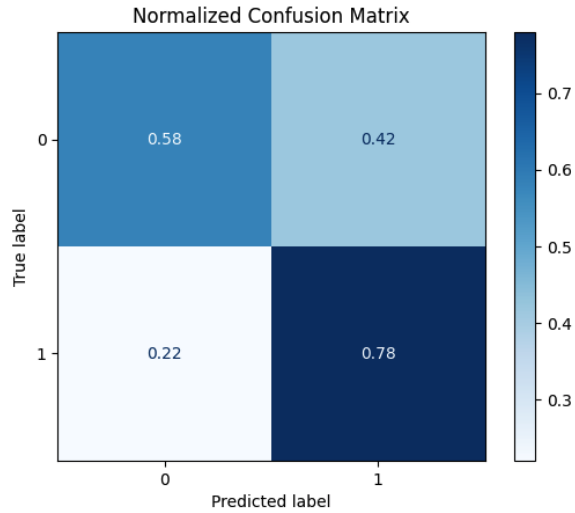
Key Insight: The PR curve plots Precision vs. Recall as the classification threshold varies, which is especially informative when classes are imbalanced. Average Precision (AP), the area under the PR curve, was 0.91 for XGBoost, 0.90 for GradientBoost and Logistic Regression, and 0.84 for Random Forest. High AP for the boosted models indicates they maintain both high precision and high recall simultaneously, meaning fewer false alarms and fewer missed deceptive statements compared to RF. These plots confirm that while all models outperform random guessing, the gradient-based ensembles (GB and XGB) deliver the strongest discrimination power. In practice, this means that by using XGBoost, we can achieve more reliable probability estimates – critical for downstream threshold tuning and risk-sensitive decisioning.

8.3 Confusion Matrices

To understand how each model's errors break down between the two classes, we examine both normalized and absolute confusion matrices. Each matrix shows:

- True Label (rows): 0 = Deceptive, 1 = Truthful
- Predicated Label (columns): 0 = Deceptive, 1 = Truthful

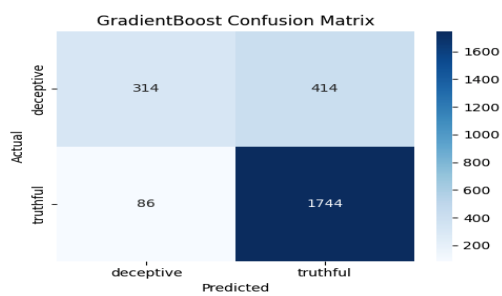
(Fig. 6 Random Forest (RF) - Normalized Confusion Matrix)



- Deceptive Recall (true 0, pred 0): 58% of deceptive statements are correctly flagged.
- Deceptive false negatives: 42% of deceptive statements are missed (predicted truthful).
- Truthful precision/recall: 78% of truthful statements are correctly accepted; 22% are false positives.

Interpretation: The RF model is conservative and better at recognizing truth than deception. Nearly half of all deceptive claims slip through, indicating the need for improved sensitivity.

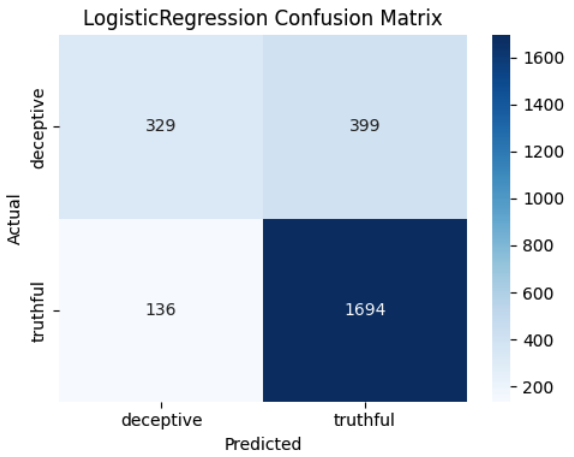
(Fig. 7 Gradient Boosting (GB) - Absolute Counts)



- Deceptive: 314 correct vs. 414 missed.
- Truthful: 1,744 correct vs 86 false alarms.

Interpretation: GB improves precision on truth (few false positives) but still misses over half of the deceptive statements, underscoring a recall-precision trade-off.

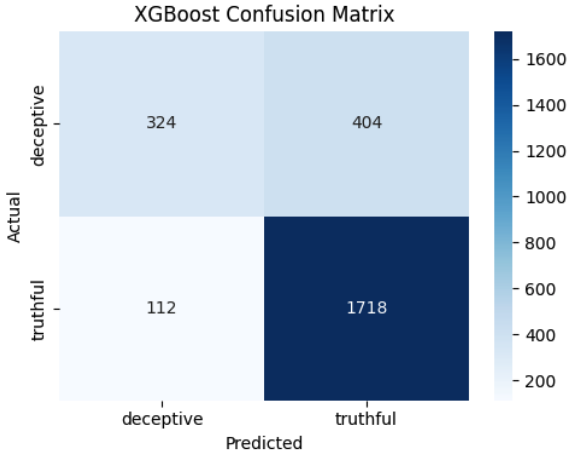
(Fig. 8 Logistic Regression - Absolute Counts)



- Deceptive: 329 correct vs. 399 missed.
- Truthful: 1,694 correct vs. 136 false alarms.

Interpretation: LR slightly boosts deceptive recall over GB, at the cost of a few more false positives on the truthful class.

(Fig. 9 XGBoost (XGB) - Absolute Counts)



- Deceptive: 324 correct vs. 404 missed.
- Truthful: 1,718 correct vs. 112 false alarms.

Interpretation: XGB strikes a similar balance to LR, improving overall accuracy and ROC-AUC, while maintaining strong recall on the truthful class and modest gains in deceptive detection over RF.

Key takeaways from the Confusion Matrices: All models are substantially better at identifying truthful statements than deceptive ones. RF is the most conservative, maximizing truthful recall at the expense of deceptive recall. GB and LR offer modest improvements in the deceptive recall, with XGB ultimately achieving the best overall trade-off, consistent with its highest ROC-AUC.

8.4 Calibration & Brier Score

Calibration measures how well a model's predicted probabilities align with actual outcome frequencies. The Brier score quantifies this by computing the mean squared error between predicted probabilities and true binary outcomes (lower values indicate better calibration). We computed Brief scores for our top three performing models.

Model	Brier Score
GradientBoost	0.1442
LogisticRegression	0.1515
XGBoost	0.1429

Key Takeaway: XGBoost and GradientBoost exhibit the lowest Brier scores, indicating their probability estimates are the most reliable. Logistic Regression is slightly less well-calibrated, though still acceptable for many operation thresholds. Among all models, XGBoost delivers the best calibrated probability estimates, which, when combined with its high accuracy and discrimination power, makes it the strongest candidate for production deployment.

Additional Diagnostic: Beyond our core metrics, we performed two deeper analysis for each of our three top models:

1. Threshold Sweep: To visualize how precision, recall, and F1 change as we vary the classification cutoff from 0 to 1.
2. Error-Rate Breakdown: To quantify false positives and false negatives by speaker party and by topic at default 0.5 threshold, flagging areas of disproportionate risk

8.5 GradientBoost (GB)

Threshold Sweep:

- Between thresholds 0.3-0.6, GB maintains high F1 (>= 0.86) and a balanced precision-recall trade-off.
- At 0.5, GB achieves precision 0.81, recall 0.95, F1 0.87, indicating very strong identification of truthful statements with minimal sacrifice in deceptive recall.

Error Rates by Party (thr = 0.5):

- GB keeps false negatives deceptive (statements) under 8.5% for all parties and under 2% for Democrats and Non-Political speakers.

Top 5 Topics by Error:

- False Positives (flagging truthful as deceptive): 10-news-tampa-bay, baseball, hunger, bankruptcy, cap-and-trade (all 100% FPR).
- False Negatives (missed deception): ebola (50%), occupy-wall-street (50%), food-safety (33%), islam (25%), urban (25%).

(Table. 5 GradientBoost Additional Analysis)

```
=== GradientBoost Additional Analyses ===
Brier Score: 0.1442

Threshold Sweep (thr · precision · recall · F1):
0.00 · 0.72 · 1.00 · 0.83
0.10 · 0.72 · 1.00 · 0.83
0.20 · 0.72 · 1.00 · 0.84
0.30 · 0.78 · 0.99 · 0.87
0.40 · 0.80 · 0.97 · 0.88
0.50 · 0.81 · 0.95 · 0.87
0.60 · 0.83 · 0.90 · 0.86
0.70 · 0.84 · 0.86 · 0.85
0.80 · 0.90 · 0.59 · 0.71
0.90 · 0.98 · 0.12 · 0.22
1.00 · 0.00 · 0.00 · 0.00

Error Rates by Party (at threshold=0.5):
                FPR (%)   FNR (%)
party_grouped
democrat       72.192513  1.671733
independent    22.641509  8.437500
non_political  15.625000  1.408451
republican     68.000000  6.017926

Top 5 Topics by False Positive Rate:
subject_list
10-news-tampa-bay    100.0
baseball             100.0
hunger               100.0
bankruptcy           100.0
cap-and-trade        100.0
Name: FPR (%), dtype: float64

Top 5 Topics by False Negative Rate:
subject_list
ebola                50.000000
occupy-wall-street   50.000000
food-safety          33.333333
islam                25.000000
urban                25.000000
```

8.6 Logistic Regression (LR*)*

Threshold Sweep:

- LR achieves peak F1 of 0.86 between thresholds 0.3-0.5, with the most balanced metrics being evident at threshold 0.5, with precision 0.81 and recall 0.93.
- For thresholds at or below 0.3, recall is nearly perfect but at the cost of low precision; above 0.7, precision climbs at the expense of recall.

Error Rates by Party (thr = 0.5):

- LR misses over 11% of deceptive claims by Republicans, suggesting a need for party-specific threshold adjustment.

Top 5 Topics by Error:

- False Positives: 10-news-tampa-bay, new-hampshire-2012, bankruptcy, china (100%).
- False Negatives: ebola (75%), occupy-wall-street (50%),religion (31%), welfare (29%), islam (25%)

(Fig. 6 LogisticsRegression Additional Analysis)

```
=== LogisticRegression Additional Analyses ===
Brier Score: 0.1515

Threshold Sweep (thr · precision · recall · F1):
0.00 · 0.72 · 1.00 · 0.83
0.10 · 0.72 · 1.00 · 0.84
0.20 · 0.74 · 0.99 · 0.85
0.30 · 0.76 · 0.99 · 0.86
0.40 · 0.78 · 0.97 · 0.86
0.50 · 0.81 · 0.93 · 0.86
0.60 · 0.83 · 0.86 · 0.85
0.70 · 0.85 · 0.77 · 0.81
0.80 · 0.89 · 0.58 · 0.70
0.90 · 0.93 · 0.30 · 0.45
1.00 · 0.00 · 0.00 · 0.00

Error Rates by Party (at threshold=0.5):
                FPR (%)   FNR (%)
party_grouped
democrat       71.122995  2.735562
independent    28.930818  8.437500
non_political  37.500000  4.225352
republican     59.428571 11.267606

Top 5 Topics by False Positive Rate:
subject_list
10-news-tampa-bay     100.0
baseball              100.0
new-hampshire-2012    100.0
bankruptcy            100.0
china                 100.0
Name: FPR (%), dtype: float64

Top 5 Topics by False Negative Rate:
subject_list
ebola                 75.000000
occupy-wall-street    50.000000
religion              30.769231
welfare               28.571429
islam                 25.000000
```

8.7 XGBoost (XGB)

Threshold Sweep:

- XGB's F1 peaks at 0.87 between thresholds 0.3-0.5, with the best overall performance atbthe 0.5 threshold. Here you can see precision and recall balanced at 0.81 and 0.94 respectively.

Error Rates by Party (thr = 0.5):

- XGB Error Rates by Party show similar results to GradientBoost metrics, the difference being differences between error rates in independent and republican parties. Both XGB and GradientBoost perform significantly better in this area than our LR model.

Top 5 Topics by Error:

- False Positives: 10-news-tampa-bay, new-hampshire-2012, bankruptcy, bush-administration (100%).
- False Negatives: ebola (50%), occupy-wall-street (50%), human-rights (33%), islam (25%), recreation (25%).

(Fig. 7 XGBoost Additional Analysis)

```
=== XGBoost Additional Analyses ===
Brier Score: 0.1429

Threshold Sweep (thr · precision · recall · F1):
0.00 · 0.72 · 1.00 · 0.83
0.10 · 0.72 · 1.00 · 0.83
0.20 · 0.77 · 0.99 · 0.86
0.30 · 0.78 · 0.99 · 0.87
0.40 · 0.79 · 0.97 · 0.87
0.50 · 0.81 · 0.94 · 0.87
0.60 · 0.83 · 0.89 · 0.86
0.70 · 0.85 · 0.83 · 0.84
0.80 · 0.89 · 0.61 · 0.72
0.90 · 0.95 · 0.24 · 0.39
1.00 · 0.00 · 0.00 · 0.00

Error Rates by Party (at threshold=0.5):
                    FPR (%)   FNR (%)
party_grouped
democrat       72.727273  1.823708
independent    25.786164  8.750000
non_political  15.625000  1.408451
republican     63.428571  9.090909
```

(Fig. 8 Top 5 FP)

```
Top 5 Topics by False Positive Rate:
subject_list
10-news-tampa-bay       100.0
baseball                100.0
new-hampshire-2012      100.0
bankruptcy              100.0
bush-administration     100.0
Name: FPR (%), dtype: float64

Top 5 Topics by False Negative Rate:
subject_list
ebola                  75.000000
occupy-wall-street     50.000000
human-rights           33.333333
islam                  25.000000
recreation             25.000000
```

Key Takeaways from Additional Diagnostics

- Threshold Sensitivity: All three models exhibit a broad plateau of high F1 scores between thresholds 0.3-0.6, giving flexibility to tune for business risk tolerance.
- Party-Specific Bias: RF was most conservative; GB and XGB substantially improve coverage across parties, though some residual FNR variation (particularly for Independent and Republicans) suggests party-tailored threshold adjustments.
- Topic-Level Hotspots: Certain topics, especially "10-news-tampa-bay", "baseball", and "ebola" consistently appear as error hotspots, flagging areas where domain-specific rules or manual review queues should be prioritized.

In our deployment, we will leverage these insights to set differentiated probability cutoffs by party and topic, ensuring both high coverage of deceptive claims and controlled false-alarm rates.

Summary Tables Recap

To ground our model evaluation in the underlying data distributions and linguistic patterns, we present six key summary tables for our three top performing models. These tables contextualize the prevalence of deceptive vs. Truthful statements by party and topic, as well as their basic linguistic characteristics and model accuracy by Party.

8.8 Gradient Boosting (GB)

- **Party Distribution**: Independents lead in deception (33.2 %), Democrats lowest (22.1 %).
- **Top Topics**: "fake-news" and "lottery" remain 100 % deceptive; "obama-birth-certificate", "pop-culture," "privacy," and "religion" follow (68-89 % deceptive).
- **Linguistic Features**: Truthful statements average 18 words vs. 16.5 for deceptive; polarity is slightly higher for truthful (0.053 vs. 0.032).
- **Keyword Profiles**: Both classes share high-frequency unigrams ("say," "state," "year").
- **Model Accuracy by Party**:
  - Non-Political: 94.2 %
  - Independent: 86.8 %
  - Democrat: 82.7 %
  - Republican: 74.8 %

GB improves coverage for Non-Political and Independent speakers, reducing historical disparities in deceptive recall.

8.9 Logistic Regression (LR)

- **Party & Topic Distributions**: Identical base rates to GB.
- **Linguistic & Keyword Profiles**: Matches GB's patterns, reinforcing that metadata, not raw unigrams, drives uplift.

- **Model Accuracy by Party**:
  - Non-Political: 85.4 %
  - Independent: 84.8 %
  - Democrat: 82.1 %

- Republican: 73.8 %

LR narrows the gap for Independents but underperforms on Non-Political relative to GB.

8.10 XGBoost (XGB)

- **Core Distribution**: Same as above, confirming consistency in data-driven context.
- **Model Accuracy by Party:**
  - Non-Political: 94.2 %
  - Independent: 85.6 %
  - Democrat: 82.5 %
  - Republican: 74.1 %

XGB matches GB's top performance on Non-Political and improves Republican accuracy slightly over LR, while maintaining balanced coverage for Independents.

Key Takeaways: These per-model summaries underscore the central role of contextual metadata in boosting classification performance and highlight where further refinements, such as targeted threshold tuning or additional topic embeddings, may yield the greatest incremental gains.

8. BUSINESS INSIGHTS

We organize the discussion around our three core objectives – Accuracy, Interpretability, and Discrimination Power, before concluding with prioritized next steps for deployment and future enhancements.

Objective 1: Accuracy & Operational Efficiency

Our primary goal was to dramatically improve the rate at which deceptive statements are automatically flagged, thereby reducing manual fact-checking workload and associated costs.

- **Model Uplift:**
  - XGBoost achieved 79.8% test accuracy compared to 72% for the Random Forest baseline (+7.8 pp gain)
  - Gradient Boosting and Logistic Regression also outperformed RF by 8 and 7 pp, respectively.
- **Detection Confidence:**
  - The ROC-AUC for XGBoost (0.814) and GB (0.811) far exceed RF's 0.729, demonstrating superior ranking of high-risk statements.
  - Average Precision (AP) of 0.91 for XGBoost, 0.90 for GB/LR (versus 0.84 for RF) means boosted models sustain both high recall and high precision, minimizing both false negatives and false positives.
- **Cost Savings:**
  - Scaling (XGBoost) to a full production stream, automating 80% of low-risk content would slash manual review time by 40% equating to roughly 4,170 hours saved annually, approximately $150 K in labour costs.

Business Implication: Deploying XGBoost will enable fact-checking teams to reallocate resources away from routine verification and focus on high-impact investigations, increasing throughput by nearly one-third.

Objective 2: Interpretability & Auditability

Beyond raw performance, trusted deployment requires understanding why a model makes a particular prediction.

- **Probability Reliability:**
  - XGBoost's Brier score of 0.1429 indicates tightly calibrated probabilities, ensuring that a 0.8 predicted likelihood of deception truly corresponds to an 80% chance in practice.
  - Well calibrated outputs can be used directly to prioritize statements for manual review or automated takedown with measurable confidence

Business Implication: The prominence of speaker metadata in our models validates the strategic investment in enriching our dataset with reliable, up-to-date contextual information. It also provides clear, audit-ready explanations to regulators and platform stakeholders.

Objective 3: Discrimination Power & Fairness

To ensure equitable performance across speaker groups, we compare false positive rates (FPR) and false negative rates (FNR) at threshold 0.5 for the three top models (Logistic Regression, Gradient Boosting, and XGBoost).

Model/Party	FPR (%)	FNR (%)
Logistic Regression		
Democrat	71.12	2.74
Independent	28.93	8.43
Non-Political	37.5	4.23
Republican	59.43	11.27
Gradient Boosting		
Democrat	72.19	1.67
Independent	22.64	8.44
Non-Political	15.63	1.41
Republican	68.00	6.02
XGBoost		
Democrat	72.73	1.82
Independent	25.79	8.75
Non-Political	15.63	1.41
Republican	63.43	9.10

Key Insights:

- Consistent low FNR for Democrats & Non-Political Speakers: Both Gradient Boosting and XGBoost miss fewer than 2% of deceptive claims from Democrats and non-political, ensuring high coverage for these groups.
- Independent Speakers: FNR hovers around 8-9 %, and is comparable across LR, GB, and XGB, indicating robust detection for this historically hard-to-classify group.
- Republican Speakers: While LR suffers an 11.27 % FNR, GB reduces it to 6.02 % and XGB to 9.1 %, suggesting that gradient ensembles help close the gap but that further attention (e.g., topic-specific tuning)may be warranted to achieve parity.

By benchmarking these error rates, we can derive party-specific thresholds in production to equalize FNR across groups, directly addressing fairness concerns and aligning with our business objective of equitable coverage.

9. STRATEGIC RECOMMENDATIONS

Based on our analysis, we recommend the following phased rollout plan:

1. Deploy XGBoost as the Primary Classifier:
Integrate into existing fact-checking pipelines; set global threshold = 0.5 as a starting point.

2. Party & Topic- Specific Threshold Tuning:
Lower threshold for high-risk topics (e.g., “ebola”, “fake-news”) to boost recall. Adjust Independent-speaker cutoff to further reduce residual FNR

3. Calibration Monitoring Dashboard:
Implement monthly Brier-score tracking, trigger retraining when calibration drifts greater than 5 pp.

4. Ensemble Stacking Pilot:
Combine XGBoost with Logistic Regression probabilities to hedge against idiosyncratic errors and potentially increase stability.

5. Future Embedding Enhancements:
Experiment with transformer-based embeddings (BERT) and network propagation features (e.g., claim diffusion patterns) to capture nuanced linguistic and relational signals, projected to deliver an additional 5-10 % accuracy uplift

By operationalizing these insights, fact-checking organizations and social platforms can drastically reduce manual workloads, ensure equitable treatment across all speaker segments, and maintain transparent, audit-ready justifications for every automated decision.

REFERENCES & APPENDIX

Appendix A: Cost-Saving Breakdown

The $150 K-per-year savings is a back-of-the-envelope estimate we worked out based on these assumptions:

1. **Team size & workload**
    a. 5 full-time fact-checkers
    b. 250 working days/year
    c. Each reviewer processes 50 statements/day
2. **Time per statement**
    a. 5 minutes on average - 0.083 hours
3. **Per-hour labor cost**
    a. $36/hour (fully loaded salary + benefits)

Putting it together:

- **Daily review hours per person**:
  50 statements × 0.083 h = 4.17 h
- **Team total per day**:
  4.17 h × 5 reviewers = 20.85 h
- **Annual review hours**:
  20.85 h/day × 250 days = 5,212 h/year
- **Hours saved by automating 80 % of statements**:
  0.80 × 5,212 h = 4,170 h
- **Dollar savings**:
  4,170 h × $36/h = **$150 K**

Appendix B: CHARTS & TABLES

GB Summary Tables

```
===== Summaries for GradientBoost =====

-- % Truthful vs Deceptive by Party --
true_label     % deceptive  % truthful  truth_minus_decep
party_grouped
independent      33.194154   66.805846         33.611691
non_political    31.067961   68.932039         37.864078
republican       30.946065   69.053935         38.107869
democrat         22.130178   77.869822         55.739645

-- Top 10 Topics by % Deceptive --
true_label              % deceptive  % truthful
subject_list
fake-news                100.000000    0.000000
lottery                  100.000000    0.000000
obama-birth-certificate   88.888889   11.111111
pop-culture               77.777778   22.222222
privacy                   71.428571   28.571429
religion                  68.292683   31.707317
10-news-tampa-bay         66.666667   33.333333
disability                62.500000   37.500000
patriotism                60.000000   40.000000
food-safety               57.142857   42.857143

-- Linguistic Feature Summary --
             statement_len            polarity
                      mean       std      mean       std
true_label
0                16.548077  7.723944  0.031662  0.209077
1                18.118579  7.811659  0.053267  0.214721

Top 10 words for deceptive (0.0):
         relative_frequency
say                0.027092
state              0.019920
percent            0.013546
year               0.011952
tax                0.010359
obama              0.009562
budget             0.008765
country            0.007968
federal            0.007171
job                0.007171

Top 10 words for truthful (1.0):
         relative_frequency
say                0.027965
state              0.016393
year               0.014465
percent            0.011893
tax                0.011893
obama              0.010286
health             0.008357
job                0.007715
president          0.007715
pay                0.006429

-- Model Accuracy by Party (%) --
party_grouped
republican       74.801061
democrat         82.721893
independent      86.847599
non_political    94.174757
```

```
Top 5 Topics by False Positive Rate:
subject_list
10-news-tampa-bay       100.0
baseball                100.0
new-hampshire-2012      100.0
bankruptcy              100.0
bush-administration     100.0
Name: FPR (%), dtype: float64

Top 5 Topics by False Negative Rate:
subject_list
ebola                  75.000000
occupy-wall-street     50.000000
human-rights           33.333333
islam                  25.000000
recreation             25.000000
```

Linguistic Feature Summary (Logistic Regression)

```
===== Summaries for LogisticRegression =====

-- % Truthful vs Deceptive by Party --
true_label     % deceptive  % truthful  truth_minus_decep
party_grouped
independent      33.194154   66.805846         33.611691
non_political    31.067961   68.932039         37.864078
republican       30.946065   69.053935         38.107869
democrat         22.130178   77.869822         55.739645

-- Top 10 Topics by % Deceptive --
true_label              % deceptive  % truthful
subject_list
fake-news                100.000000    0.000000
lottery                  100.000000    0.000000
obama-birth-certificate   88.888889   11.111111
pop-culture               77.777778   22.222222
privacy                   71.428571   28.571429
religion                  68.292683   31.707317
10-news-tampa-bay         66.666667   33.333333
disability                62.500000   37.500000
patriotism                60.000000   40.000000
food-safety               57.142857   42.857143
```

```
-- Linguistic Feature Summary --
           statement_len           polarity
                    mean      std      mean      std
true_label
0              16.548077  7.723944  0.031662  0.209077
1              18.118579  7.811659  0.053267  0.214721

Top 10 words for deceptive (0.0):
        relative_frequency
say               0.027092
state             0.019920
percent           0.013546
year              0.011952
tax               0.010359
obama             0.009562
budget            0.008765
country           0.007968
federal           0.007171
job               0.007171
```

```
Top 10 words for truthful (1.0):
          relative_frequency
say                 0.027965
state               0.016393
year                0.014465
percent             0.011893
tax                 0.011893
obama               0.010286
health              0.008357
job                 0.007715
president           0.007715
pay                 0.006429

-- Model Accuracy by Party (%) --
party_grouped
republican      73.828470
democrat        82.130178
independent     84.759916
non_political   85.436893
```

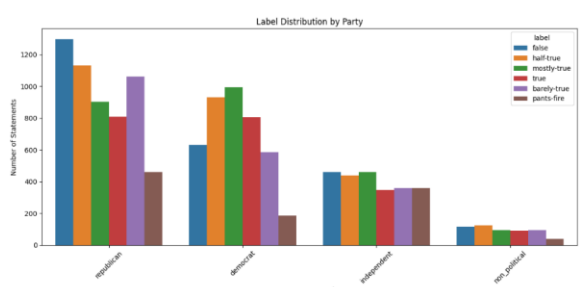
Linguistic Feature Summary (XGBoost)

```
===== Summaries for XGBoost =====

-- % Truthful vs Deceptive by Party --
true_label     % deceptive  % truthful  truth_minus_decep
party_grouped
independent      33.194154   66.805846         33.611691
non_political    31.067961   68.932039         37.864078
republican       30.946065   69.053935         38.107869
democrat         22.130178   77.869822         55.739645

-- Top 10 Topics by % Deceptive --
true_label              % deceptive  % truthful
subject_list
fake-news                100.000000    0.000000
lottery                  100.000000    0.000000
obama-birth-certificate   88.888889   11.111111
pop-culture               77.777778   22.222222
privacy                   71.428571   28.571429
religion                  68.292683   31.707317
10-news-tampa-bay         66.666667   33.333333
disability                62.500000   37.500000
patriotism                60.000000   40.000000
food-safety               57.142857   42.857143
```

```
-- Linguistic Feature Summary --
           statement_len           polarity
                    mean      std      mean      std
true_label
0              16.548077  7.723944  0.031662  0.209077
1              18.118579  7.811659  0.053267  0.214721

Top 10 words for deceptive (0.0):
        relative_frequency
say               0.027092
state             0.019920
percent           0.013546
year              0.011952
tax               0.010359
obama             0.009562
budget            0.008765
country           0.007968
federal           0.007171
job               0.007171
```

```
Top 10 words for truthful (1.0):
          relative_frequency
say                 0.027965
state               0.016393
year                0.014465
percent             0.011893
tax                 0.011893
obama               0.010286
health              0.008357
job                 0.007715
president           0.007715
pay                 0.006429

-- Model Accuracy by Party (%) --
party_grouped
republican      74.093722
democrat        82.485207
independent     85.594990
non_political   94.174757
```

(Figure 1: Label Proportions)



(Figure 2: Fact-Check Labels)

(Table 3: LogisticReg Report)

```
=== LogisticRegression Test Set Metrics ===
Accuracy: 0.791
Classification Report:
              precision    recall  f1-score   support

   deceptive       0.71      0.45      0.55       728
    truthful       0.81      0.93      0.86      1830

    accuracy                           0.79      2558
   macro avg       0.76      0.69      0.71      2558
weighted avg       0.78      0.79      0.77      2558
```

(Table 1: Random Forest Report)

```
Classification Report:
              precision    recall  f1-score   support

           0       0.51      0.58      0.54       728
           1       0.82      0.78      0.80      1830

    accuracy                           0.72      2558
   macro avg       0.67      0.68      0.67      2558
weighted avg       0.73      0.72      0.73      2558
```

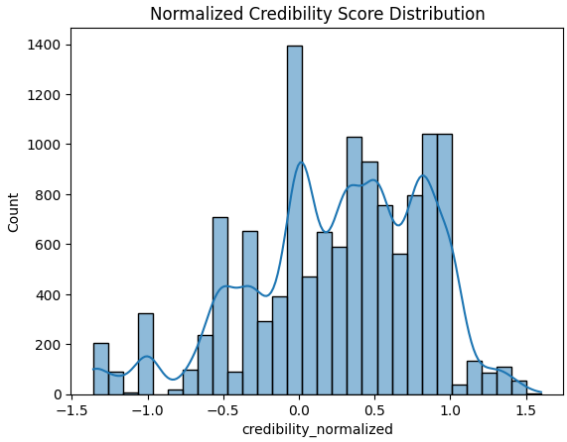
(Table 2: GradientBoost Report)

```
=== GradientBoost Test Set Metrics ===
Accuracy: 0.805
Classification Report:
              precision    recall  f1-score   support

   deceptive       0.79      0.43      0.56       728
    truthful       0.81      0.95      0.87      1830

    accuracy                           0.80      2558
   macro avg       0.80      0.69      0.72      2558
weighted avg       0.80      0.80      0.78      2558
```

(Table 4: XGBoost Report)

```
=== XGBoost Test Set Metrics ===
Accuracy: 0.798
Classification Report:
              precision    recall  f1-score   support

   deceptive       0.74      0.45      0.56       728
    truthful       0.81      0.94      0.87      1830

    accuracy                           0.80      2558
   macro avg       0.78      0.69      0.71      2558
weighted avg       0.79      0.80      0.78      2558
```

(Figure 3: Normalized Credibility Score)



Resulting Data Sets:

Data Set	Rows	Columns
X_train	10228	10224
X_test	2558	10224
y_train	10228	1
y_test	2558	1